TOURNAMENT SCHEDULER

COMBINATORICS & GRAPH THEORY

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TOURNAMENT SCHEDULER?

Systematically organizing matches among participants to ensure a smooth, balanced, and efficient competition.

Fair Play

Conflict-Free

Optimized

Enhanced

01. COMBINATORICS

Counting and arranging competition dynamics.

TOTAL POSSIBLE MATCHES

$$\genfrac{}{}{0pt}{}{n}{2} = \frac{n(n-1)}{2}$$

Example: 4 Teams (A, B, C, D)

Matches: AB, AC, AD, BC, BD, CD

TOTAL: 6

TEAM ARRANGEMENTS

24

POSSIBLE WAYS

Arranging Teams ($n!$)

Changing team positions changes future matchups and bracket flows.

$$4! = 4 \times 3 \times 2 \times 1 = 24$$

SCHEDULING PERMUTATIONS

Schedule A (Standard)

R1	AB , CD
R2	AC , BD
R3	AD , BC

Schedule B (Optimized)

R1	AC , BD
R2	AB , CD
R3	AD , BC

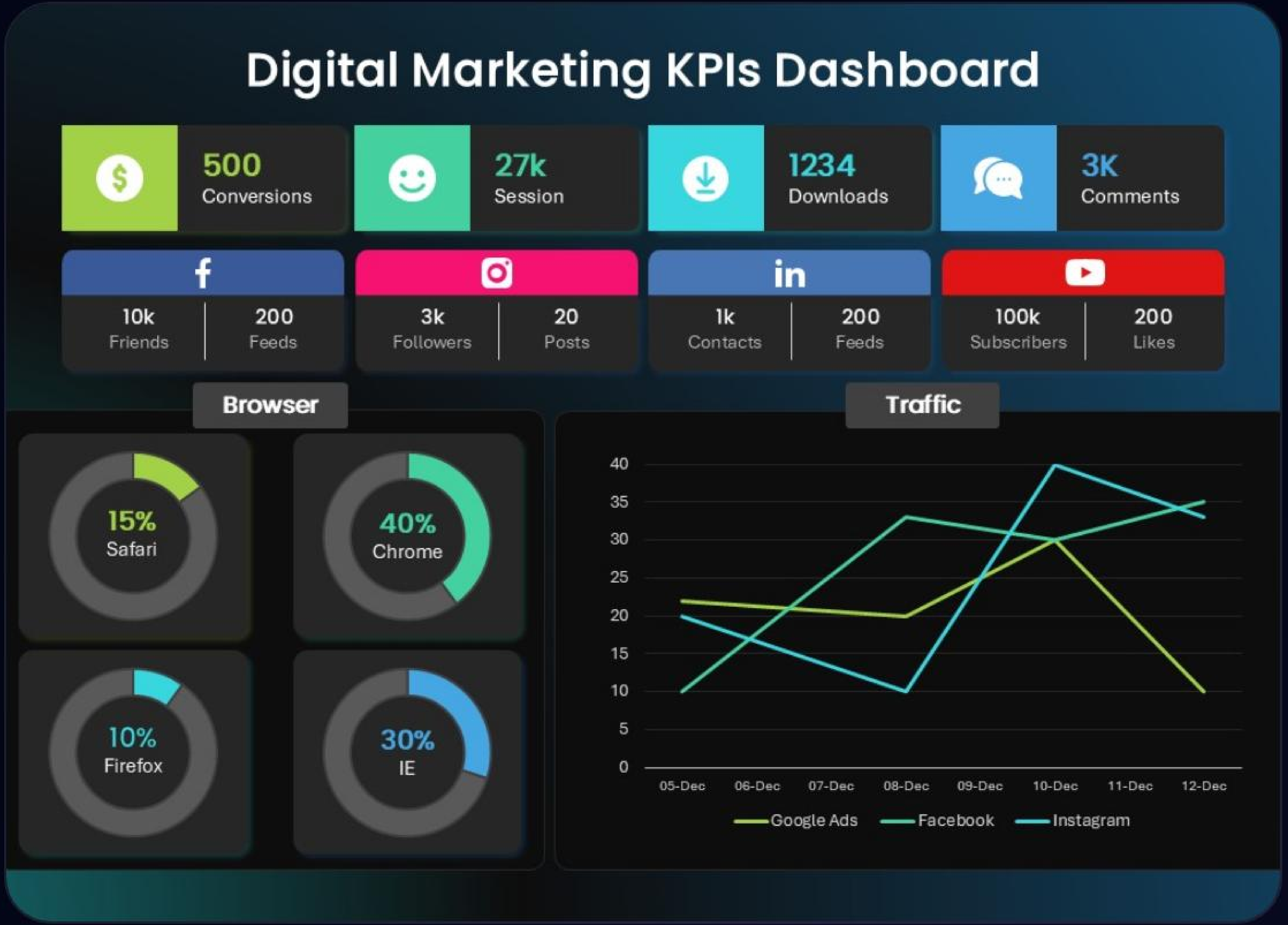
Permutations change the **timing/order** of rounds even with the same matches.

02. GRAPH THEORY

Representing relationships using nodes and edges.

VISUALIZING MATCHUPS

- **Nodes:** Representing teams or participants.
- **Edges:** Representing a match between participants.
- **Structure:** Graph theory models the entire bracket topology.



SINGLE ELIMINATION

One loss and you are out. Most common for high-stakes competition.

Matches (m): $n - 1$
Participants: 2^i

DOUBLE ELIMINATION

Second chances. Move to the loser's bracket after the first loss.

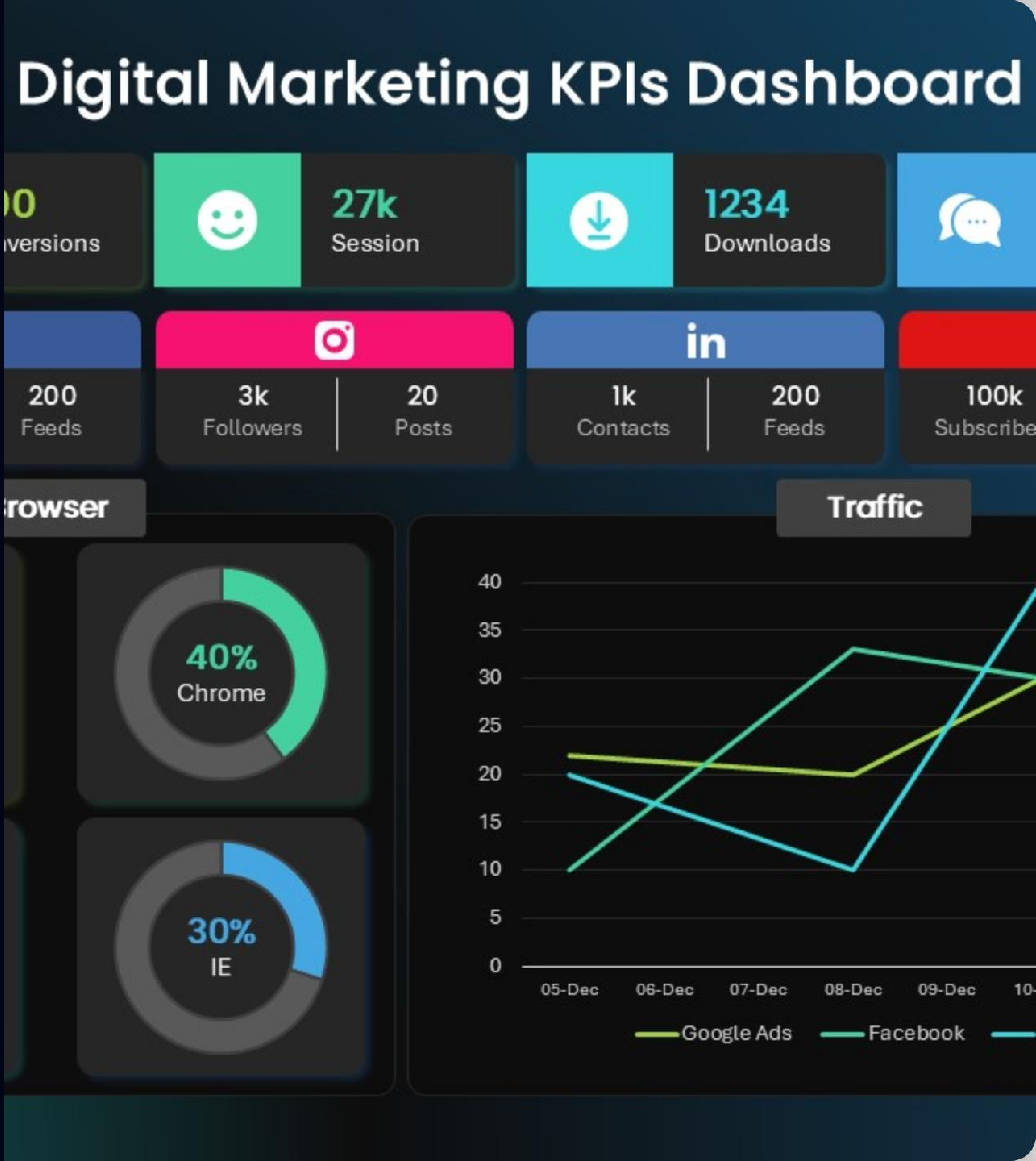
Matches (m): $2n - 2$

For 8 teams, there are 14 matches total.

ROUND ROBIN

Total fairness. Every participant competes against all others.

$$\text{Total Games} = \frac{n(n-1)}{2}$$



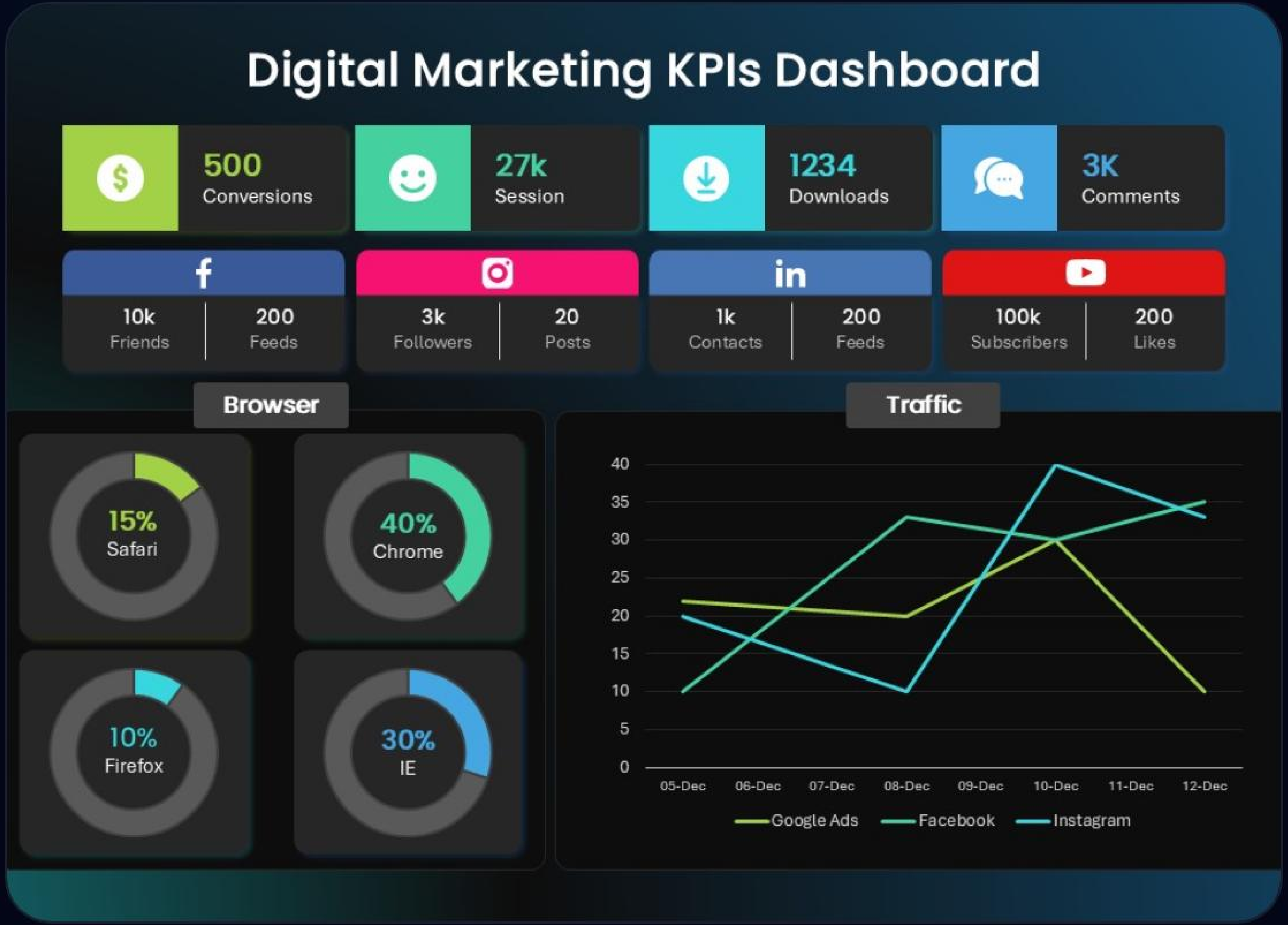
THE POLYGON METHOD

Polygon Method Diagram

Systematic Pairing

1. Draw a polygon with $N-1$ vertices.
2. Place one team in the center.
3. Connect vertices across the polygon.
4. Rotate the polygon clockwise for each round.

ANALYSIS: COMPLETE GRAPHS (K_N)



K_8 Analysis (8 Teams)

In a Round Robin with 8 teams, every team is a vertex in a complete graph K_8.

Vertices: 8
Edges: $8C2 = 28$ Matches
Degree of each Node: 7

Every team plays 7 games, representing the edges connected to their node.

THANK YOU!

ANY QUESTIONS?

FIFA / PSL
Real World Impact

SWISS / LADDER
Alternative Formats

GROUP 19
Tournament Scheduler

IMAGE SOURCES

https://img.magnific.com/free-photo/empty-dark-room-modern-futuristic-sci-fi-background-3d-illustration_35913-2349.jpg?semt=ais_hybrid&w=740&q=80

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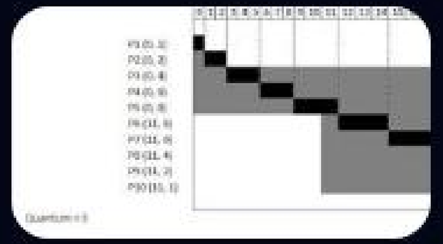

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Source: en.wikipedia.org